

contrastive loss

$$L = (1 - Y) \cdot D^2 + Y \cdot \max(0, m - D)^2$$

$$Y = 0$$

(positive pair)

similar images



$$L = (1 - 0) \cdot D^2$$

$$L = D^2$$

n/w tries to reduce D to make embeddings closer.

D = euclidean distance
betn embeddings

If D is small $\Rightarrow$
loss becomes small
Goal

Why square the D $\Rightarrow$

if dist is large $\rightarrow$ penalty becomes bigger. This helps model learn that similar images should be close.

$$Y = 1$$

(negative pair)

different images



$$L = \max(0, m - D)^2$$

Here, loss fn penalizes small dist & encourages D to be at least m .

$m \Rightarrow$ margin $\Rightarrow$ threshold that enforces a minimum distance for dissimilar pairs

[increases the separation betn similar & dissimilar vector]

- If D is small $\rightarrow$ loss is high $\rightarrow$ forcing n/w to increase D .
- If D is already greater than $m \rightarrow$ loss becomes zero = meaning no further pushing needed.

given $Y = 0$
 $D = 0.8$

$$\begin{aligned} L &= (1 - 0) \cdot D^2 \\ &= (1 - 0) \cdot (0.8)^2 \\ &= 0.64. \end{aligned}$$

The loss encourages
 η/w to further reduce
 D to 0, making
embeddings closer.

given $Y = 1$
 $D = 1.2$ $M = 2$

$$\begin{aligned} L &= \max(0, M - D)^2 \\ &= \max(0, 2 - 1.2)^2 \\ &= (0.8)^2 \\ &= 0.64. \end{aligned}$$

As D is 1.2 which is
less than margin (2)
Loss is nonzero,
forcing η/w to increase
the distance betⁿ
embeddings.